CS204: Algorithms & Data Structures Lab

Week # 02 (1 Questions, 100 Marks)
Lab session: AL1
Held on: 05-Aug-2024 (Mon)
Lab Timings: 14:00 to 17:00 Hours   Pages: 3
Submission time: 16:45 Hrs, 05-Aug-2024
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1. This lab theme is based on tutorial 01
2. You can use one C++ text book of your choice
3. Any use of advanced concepts in C++ other than the ones discussed in tutorial will invite `0 marks`.
4. You must write the program in one single file whose name should be `your_roll_number.cc`. If your roll number is 240101999 then the file name should be `240101999.cc`

Question 1: (100 points)
Use C++ related features only and implement the following

Input file Description. The input file `curriculum.txt` contains 64 lines. Except for the first line, all the remaining lines have a fixed structure as described below.

1. First line `63` represents the number of lines that follows first line.
2. From second line onwards, every line has `8` columns
3. First column represents the curriculum corresponding to the department
4. Second column represents semester number
5. Third column represents course id
6. Fourth column represents course name
7. Fifth column represents number of lecture hours per week
8. Sixth column represents number of tutorial hours per week
9. Seventh column represents number of practical hours per week
10. Eighth column represents assigned number of credits

Task 01 (8 marks) Variable declaration. Declare the following arrays statically

1. (1 mark) A two-dimensional `char` array of size `lines` $\times$ 3 to hold department name
2. (1 mark) An `int` array of size `lines` to hold semester number.
3. (1 mark) A two-dimensional `char` array of size `lines` $\times$ 6 to hold course id.
4. (1 mark) A two-dimensional `char` array of size `lines` $\times$ 80 to hold course name.
5. (1 mark) An `int` array of size `lines` to hold number of lecture hours.

6. (1 mark) An `int` array of size `lines` to hold number of tutorial hours
7. (1 mark) An `int` array of size `lines` to hold number of practical hours
8. (1 mark) An `int` array of size `lines` to hold number of credits

Task 02 (10 marks) Reading data from input file `curriculum.txt`. Read the first line into an `int` variable say `lines`. For each of the line and read data in each column into respective variables declared above file `curriculum.txt`. Use indirection operation to read data from the file. Discuss related to indirection is described in the tutorial lecture slides (slide # 13). You of FILE I/O is NOT allowed.

Task 03 (8 marks) Compute the following

1. (2 marks) Total number of lecture hours (for all the course)
2. (2 marks) Total number of tutorial hours (for all the course)
3. (2 marks) Total number of practical hours (for all the course)
4. (2 marks) Total number of credits (for all the course)

Task 04 (8 marks) Compute the following

1. (2 marks) Total number of lecture hours which are offered in even semester (for all the course)
2. (2 marks) Total number of tutorial hours which are offered in even semester (for all the course)
3. (2 marks) Total number of practical hours which are offered in even semester (for all the course)
4. (2 marks) Total number of credits which are offered in even semester (for all the course)

Task 05 (8 marks) Compute the following

1. (2 marks) Total number of lecture hours which are offered by CSE department (for all the course)
2. (2 marks) Total number of tutorial hours which are offered by CSE department (for all the course)
3. (2 marks) Total number of practical hours which are offered by CSE department (for all the course)
4. (2 marks) Total number of credits which are offered by CSE department (for all the course)

Task 06 (19 marks) Perform the following

1. (8 marks) Modify the task 01 by creating dynamic memory for each of the eight described arrays.
2. (2 marks) Copy the data present in eight array from task 02 to the dynamically created arrays.
3. (8 marks) Repeat tasks 03 to 05.
4. (1 marks) Remove all the dynamically allocated memory

Task 07 (39 marks) Perform the following

1. (4 marks) Create dynamic memory for one-dimensional `int` arrays `lectures, tutorials, practicals` and `credits`.
2. (4 marks) Copy the corresponding data to these four arrays.
3. (1 mark) Declare four `int` pointers say p_1, p_2, $p_3 and p_4$
4. (1 mark) Assign the address of 30^{th} array location of `lectures` to p_1
5. (1 mark) Assign the address of 30^{th} array location of `tutorials` to p_2
6. (1 mark) Assign the address of 30^{th} array location of `practicals` to p_3
7. (1 mark) Assign the address of 30^{th} array location of `credits` to p_4
8. Using the pointers p_1, p_2, p_3 and p_4 compute
   (a) (8 marks) quantities in task 03 by using pointer arithmetic
   (b) (8 marks) quantities in task 04 by using pointer arithmetic
   (c) (8 marks) quantities in task 05 by using pointer arithmetic
9. (2 marks) Remove all the dynamically allocated memory.